

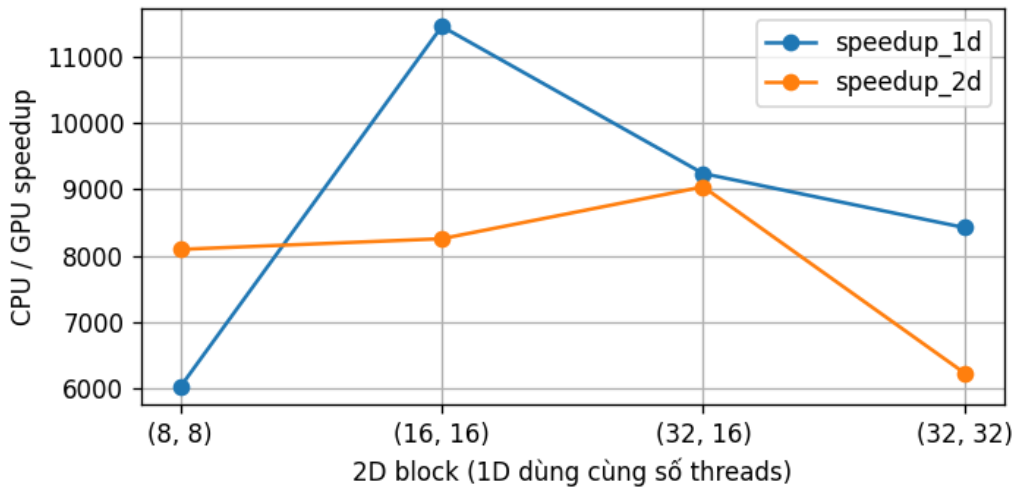
Lab 4: Threads

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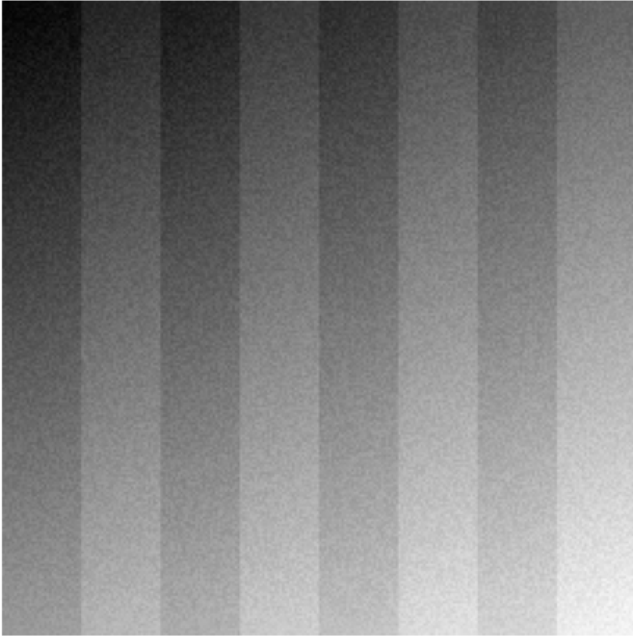
The 2D kernel maps thread (x, y) to pixel (y, x) . The notebook compares 1D and 2D blocks with the same number of threads and plots CPU/GPU speedup. Exercise 1: 16×16 permits four blocks and 1024 threads/SM; 8×8 permits only 512 threads with eight blocks; 32×32 exceeds 512 threads/block. Exercise 2: 128, 256, 512 and 1024 threads/block give 512, 1024, 1536 and 1024 resident threads respectively. Choose 512. Exercise 3: $\lceil 2000/512 \rceil 512 = 2048$ launched threads; 48 are inactive. These exercise limits are hypothetical; benchmarks use the actual GPU limits.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.

block	cpu_ms	gpu_1d_ms	gpu_2d_ms	speedup_1d	speedup_2d	max_error
(8, 8)	463.127	0.0768503	0.0572205	6026.35	8093.73	1
(16, 16)	463.127	0.0404517	0.0561078	11448.9	8254.22	1
(32, 16)	463.127	0.0501474	0.05126	9235.31	9034.86	1
(32, 32)	463.127	0.0549952	0.0743071	8421.22	6232.6	1



CPU



GPU 2D

